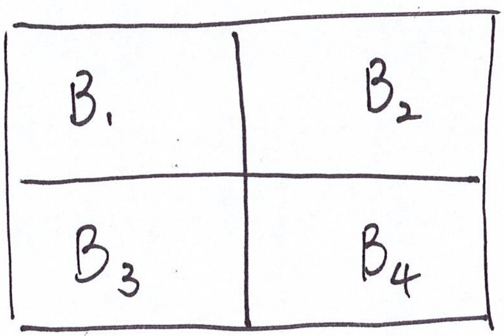
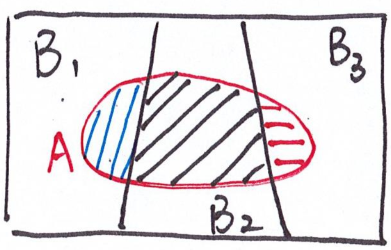
Rule of average conditional probabilities

▽ Def. If $\bigcup_{j=1}^{n} B_j = \Omega$ and B_j's are pairwise disjoint, then B_j's partition Ω.



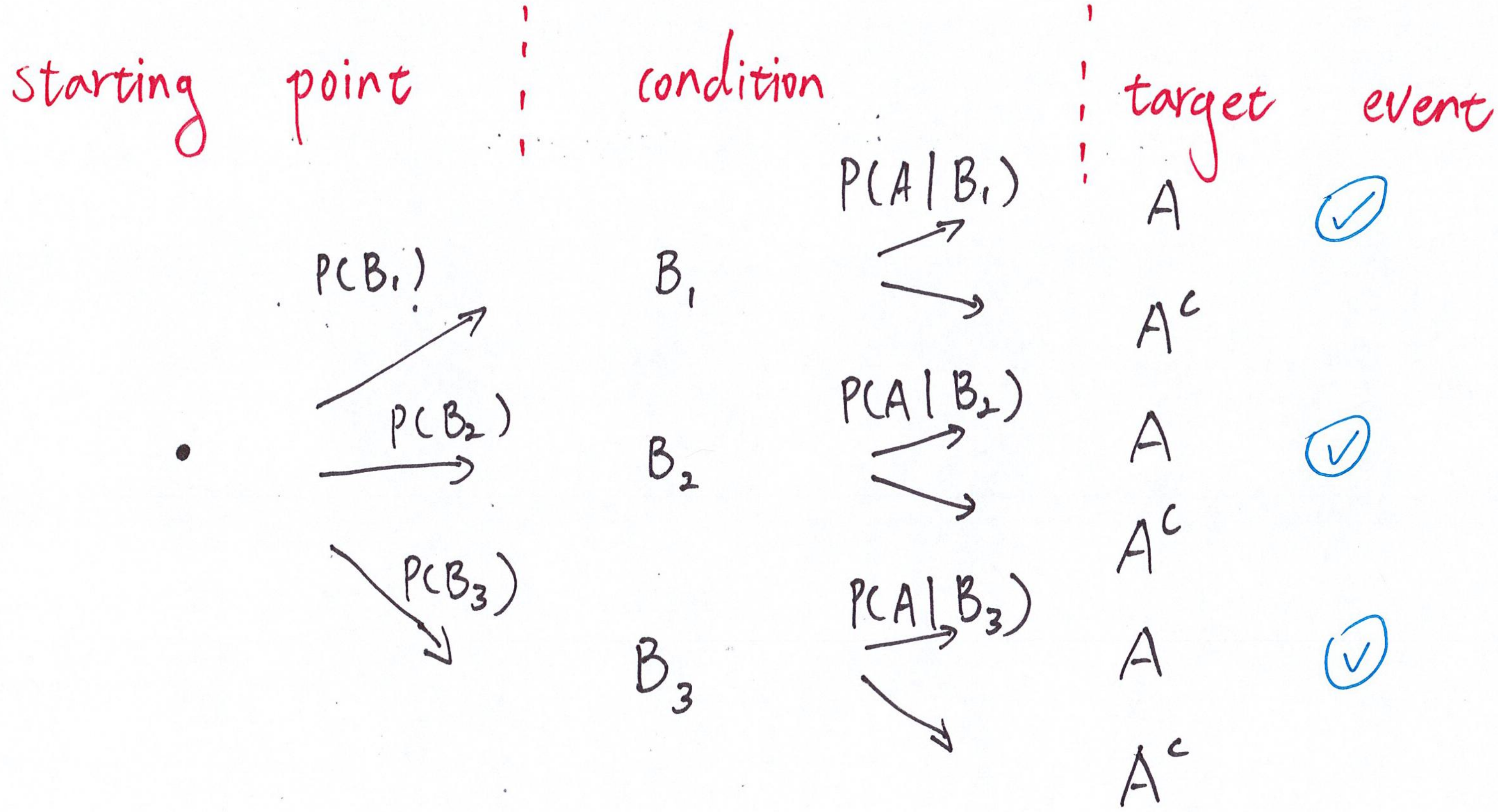
▽ Consider



$$P(A) = P(A \cap B_1) + P(A \cap B_2) + P(A \cap B_3)$$

$$= P(A \mid B_1) \cdot P(B_1) + P(A \mid B_2) \cdot P(B_2) + P(A \mid B_3) \cdot P(B_3)$$

Tree diagram : Each branch represents the evolution of an experiment.



$$P(A) = P(B_1) \cdot P(A|B_1) + P(B_2) \cdot P(A|B_2) + P(B_3) \cdot P(A|B_3)$$

Ex. There are 2 dice in a hat. One die is 6-sided, and the other one is 20-sided. You randomly pick one die and roll it.

Let $B = \{\text{the result} = 5\}$.

Find $P(B)$.

Let $A = \{\text{the 20-sided is picked}\}$

$$
\begin{tikzcd}
 & & B \checkmark & P(A)\cdot P(B|A) \\
 & A \arrow[ur, "P(B|A)"] \arrow[dr] & & \\
 & & B^c & \\
\bullet \arrow[uur, "P(A)"] \arrow[ddr, "P(A^c)"'] & & & \\
 & & B \checkmark & P(A^c)\cdot P(B|A^c) \\
 & A^c \arrow[ur, "P(B|A^c)"] \arrow[dr] & & \\
 & & B^c &
\end{tikzcd}
$$

$$P(B) = \frac{1}{2}\cdot\frac{1}{20} + \frac{1}{2}\cdot\frac{1}{6} = \frac{13}{120}$$

☞ Baye's theorem : • $P(A \cap B_1) = P(A \mid B_1) \cdot P(B_1)$

• $P(A) = \sum_{j=1}^{n} P(A \mid B_j) \cdot P(B_j)$

by the rule of average conditional probabilities

Collectively, we have

$$P(B_1 \mid A) = \frac{P(A \cap B_1)}{P(A)}$$

$$= \frac{P(A \mid B_1) \cdot P(B_1)}{\sum_{j=1}^{n} P(A \mid B_j) \cdot P(B_j)}$$

▷ Ex. Suppose that there are two types of drivers:

- Type I drivers submit at least one insurance claim per year with probability 0.1.

- Type II ... ... with probability 0.3.

Moreover, 80% of the population are type I drivers.

Given that a driver submitted a claim, find the probability that the driver is of type II.

Let $A = \{$ the selected driver is a type II driver $\}$

$B = \{$ a driver submitted a claim $\}$

We know

- $P(A) = 1 - 0.8 = 0.2$
- $P(B \mid A^c) = 0.1$
- $P(B \mid A) = 0.3$

By Baye's theorem,

$$P(A \mid B) = \frac{P(B \mid A) \cdot P(A)}{P(B \mid A) \cdot P(A) + P(B \mid A^c) \cdot P(A^c)}$$

$$= \frac{0.3 \times 0.2}{0.3 \times 0.2 + 0.1 \times 0.8} = \frac{3}{7}.$$